

Assignment 2: AWS Lambda layer for Matrix Transposition

Developing the Transpose function

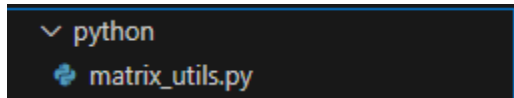
- a. Created a python function to take input as matrix and return transpose of matrix

```
python > matrix_utils.py > ...  
1  def transpose(mat):  
2      m,n = len(mat),len(mat[0])  
3      imat = [[0 for i in range(m)] for j in range(n)]  
4      for i in range(m):  
5          for j in range(n):  
6              imat[j][i] = mat[i][j]  
7      return imat  
8
```

- b.

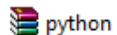
Creating and deploying a lambda layer

- a. Added the file 'matrix_utils.py' in folder named 'python'.



- b.

- c. Zip the folder.



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WinRAR ZIP archive

1 KB

- d.

- e. In AWS go to lambda-> layers and Create a new layer.

- f. Upload the zipped python folder and select other options accordingly.

aws Services Search [Alt+S]

Create layer

Layer configuration

Name
matrix_layer

Description - optional
Transpose matrix

☒ Upload a .zip file
☐ Upload a file from Amazon S3

 Upload

python.zip
0.25 KB

For files larger than 10 MB, consider uploading using Amazon S3.

Compatible architectures - optional [Info](#)
Choose the compatible instruction set architectures for your layer.

☒ x86_64
☒ arm64

Compatible runtimes - optional [Info](#)
Choose up to 15 runtimes.

Runtimes

Python 3.11

License - optional [Info](#)

g.

Utilizing Lambda layer in a function.

a. Created a function 'gds-lambda-assign-2'.

aws Services Search [Alt+S]

Lambda > Functions > gds-lambda-assign-2

gds-lambda-assign-2

Throttle Copy ARN Actions

Export to Application Composer Download

Function overview [Info](#)

Diagram Template

 gds-lambda-assign-2

Layers (1)

+ Add trigger

+ Add destination

Description
-

Last modified
8 minutes ago

Function ARN
arn:aws:lambda:us-east-1:533267408419:function:gds-lambda-assign-2

Function URL [Info](#)
-

b.

c. Add the created layer in the layers section.

d. Select the 'Custom layers'

[Lambda](#) > [Layers](#) > Add layer

Add layer

Function runtime settings

Runtime Python 3.11	Architecture x86_64
------------------------	------------------------

Choose a layer

Layer source [Info](#)
Choose from layers with a compatible runtime and instruction set architecture or specify the Amazon Resource Name (ARN) of a layer version. You can also [create a new layer](#).

☐ AWS layers
Choose a layer from a list of layers provided by AWS.

☒ Custom layers
Choose a layer from a list of layers created by your AWS account or organization.

☐ Specify an ARN
Specify a layer by providing the ARN.

Custom layers

Layers created by your AWS account or organization that are compatible with your function's runtime.

matrix_layer ▼

Version
1 ▼

[Cancel](#) [Add](#)

e.

Layers Info						Edit	Add a layer
Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN		
1	matrix_layer	1	python3.11	x86_64, arm64	arn:aws:lambda:us-east-1:533267408419:layer:matrix_layer:1		

f.

g. Write code and configure 'Test' to execute the code.

lambda_function × Execution results × Environment Vari × +

```
1 import json
2 from matrix_utils import transpose
3
4 def lambda_handler(event, context):
5     print(transpose([[1,2,3],[4,5,6]]))
6     return {
7         'statusCode': 200,
8         'body': json.dumps('Hello from Lambda!')
9     }
10
```

h.

Response	
{ "statusCode": 200, "body": "\"Hello from Lambda!\"" }	
Function Logs	
START RequestId: 57ebe197-feb4-4cfe-ac4d-98ec0f42b1e9 Version: \$LATEST [[1, 4], [2, 5], [3, 6]]	
END RequestId: 57ebe197-feb4-4cfe-ac4d-98ec0f42b1e9	
REPORT RequestId: 57ebe197-feb4-4cfe-ac4d-98ec0f42b1e9 Duration: 1.52 ms Billed Duration: 2 ms Memory Size: 128 MB Max Memory Used: 40 MB	
Request ID	
57ebe197-feb4-4cfe-ac4d-98ec0f42b1e9	

i.